

Education

University of Toronto

BS, Major in Computer Science and Mathematics

Sep. 2023 - Present | Toronto, CA

Expected Graduation: Jun. 2028

CGPA: 3.7 / 4.0

Tsinghua University

Exchange, Computer Science

Sep. 2025 - Present | Beijing, CN

Links

Blog: alittlebear.net

Github:// [josephdsiu](https://github.com/josephdsiu)

LinkedIn:// [a-little-bear](https://www.linkedin.com/in/a-little-bear)

Coursework

AI & Machine Learning

Machine Learning (Prof. Jie Tang & Jun Zhu)

Artificial Intelligence (Prof. Qingyao Ai)

Foundations of Artificial Intelligence (Prof.

Huaping Liu & Fuchun Sun)

Computer Vision (Prof. Junliang Xing)

Introduction to Machine Learning

Introduction to Deep Learning (Prof.

Xiaolin Hu)

Logic & Theory

Comp. Complexity and Computability (A+)

Introduction to Mathematical Logic (A+)

Enriched Theory of Computation (A+)

Enriched Data Structures and Analysis (A-)

Algorithm Design and Analysis (A+)

Analysis I / II (A+)

Systems & Graphics

Principles and Practice of Compiler

Construction

Fundamentals of Computer Graphics (Prof.

Yong-Jin Liu)

Skills

Languages

Java • Python • C/C++ • $\LaTeX$

Mandarin (Native) • English

(Native/Bilingual)

Frameworks & Tools

PyTorch • NumPy • Scikit-Learn • Pandas •

Git • Linux

Research Interests

AGI & Cognitive Architectures

- **Reasoning in LMMs:** Enhancing System 2 reasoning via MCTS and CoT; integrating Causal Inference with Neuro-Symbolic systems to revive expert-level logic.
- **Mechanistic Interpretability:** Bridging Brain-Model Alignment; analyzing Superposition and Sparsity to decipher knowledge representation.

Generalist Agents & Self-Evolution

- **Zero-Shot Generalization:** Developing Generalist Agents capable of adapting to unseen environments without fine-tuning (In-Context RL).
- **Collective Intelligence & Auto-Science:** Investigating emergent behaviors in Collective Intelligence; applying Meta-learning for Automated Scientific Discovery (Agentic Science).

Research and Academic Projects

Rethinking Manifold Assumptions in Diffusion Models

Dec 2025 – Dec 2025

- Challenged the linear embedding assumption in "Just image Transformers" (JiT) by introducing Non-linear Patch Embeddings (SiLU).
- Key Insight: Analyzed the Precision-Recall Trade-off in backbone design—proving that compression (Bottlenecks) favors fidelity (High Precision) while capacity (SviGLU) favors diversity (High Recall).

Ego-GAT-SqueezeNet: Multi-Agent Social Behavior Understanding

Nov 2025 – Jan 2026

- Proposed Ego-GAT-SqueezeNet, unifying Egocentric Alignment and Graph Attention Networks (GAT) to model invariant agent-target topology from raw skeletal data.
- Integrated a Squeezeformer backbone to efficiently capture long-range temporal dependencies and designed a Lab Context Adapter using FiLM (Feature-wise Linear Modulation) to mitigate cross-domain shifts.
- Solved extreme class imbalance (>76% background) using Action-Rich Sampling and Dataset-aware Focal Loss.

PastureNet: Cross-Domain Biomass Estimation via Foundation Models

Oct 2025 – Jan 2026

- Developed PastureNet, a hierarchical ensemble framework integrating DINOv3, SigLIP 2, and EVA-02 to capture object-centric, semantic, and texture-sensitive features respectively.
- Innovated Zero-shot Semantic Concept Scores (e.g., "clover presence") using SigLIP 2 to inject explicit ecological domain knowledge directly into the regression pipeline.
- Implemented Matrix Reconciliation for post-processing to enforce biological consistency across hierarchical targets.